

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **Perkadox CH-50**  
 Product Description: **Perkadox CH-50**  
 Cat No. : 630140000  
 Molecular Formula C14 H10 O4

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**E-mail address**

Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                   |                     |
|-----------------------------------|---------------------|
| Organic peroxides                 | Type D (H242)       |
| Serious Eye Damage/Eye Irritation | Category 2 (H319)   |
| Skin Sensitization                | Category 1 (H317)   |
| Reproductive Toxicity             | Category 1B (H360D) |
| Acute aquatic toxicity            | Category 1 (H400)   |
| Chronic aquatic toxicity          | Category 1 (H410)   |

**Label Elements**


Signal Word

Danger

# SAFETY DATA SHEET

Perkadox CH-50

Revision Date 22-Mar-2025

## Hazard Statements

H242 - Heating may cause a fire  
H317 - May cause an allergic skin reaction  
H319 - Causes serious eye irritation  
H410 - Very toxic to aquatic life with long lasting effects  
H360D - May damage the unborn child

## Precautionary Statements

### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P220 - Keep away from clothing and other combustible materials  
P234 - Keep only in original packaging  
P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P410 - Protect from sunlight  
P411 + P235 - Store at temperatures not exceeding 25 °C/ 77 °F. Keep cool  
P420 - Store away from other materials

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Contains a known or suspected endocrine disruptor  
Included in the list established in accordance with Article 59(1) for having endocrine disrupting properties  
Contains a substance on the National Authorities Endocrine Disruptor Lists

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                        | CAS No  | Weight % |
|--------------------------------------------------|---------|----------|
| Dibenzoyl peroxide                               | 94-36-0 | 50-60    |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 84-61-7 | 40-50    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

# SAFETY DATA SHEET

Perkadox CH-50

Revision Date 22-Mar-2025

|                                                                                                                                                                                                                          |                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ingestion</b>                                                                                                                                                                                                         | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| <b>Inhalation</b>                                                                                                                                                                                                        | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b>                                                                                                                                                                                | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| <b><u>Most important symptoms and effects, both acute and delayed</u></b>                                                                                                                                                |                                                                                                                                                  |
| May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing. |                                                                                                                                                  |
| <b><u>Indication of any immediate medical attention and special treatment needed</u></b>                                                                                                                                 |                                                                                                                                                  |
| <b>Notes to Physician</b>                                                                                                                                                                                                | Treat symptomatically.                                                                                                                           |

## SECTION 5: FIREFIGHTING MEASURES

### **Extinguishing media**

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Special hazards arising from the substance or mixture**

Do not allow run-off from fire-fighting to enter drains or water courses. Dust can form an explosive mixture with air. Organic peroxide. May undergo hazardous decomposition. These substances will accelerate burning when involved in a fire.

#### **Hazardous Combustion Products**

benzoic acid, Carbon dioxide (CO<sub>2</sub>).

### **Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### **Personal Precautions, Protective Equipment and Emergency Procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

### **Environmental precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### **Methods and Material for Containment and Cleaning Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

# SAFETY DATA SHEET

Perkadox CH-50

Revision Date 22-Mar-2025

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep away from combustible material. Keep at temperatures below 25°C. Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component          | Malaysia | ACGIH TLV                | OSHA PEL                                                       |
|--------------------|----------|--------------------------|----------------------------------------------------------------|
| Dibenzoyl peroxide |          | TWA: 5 mg/m <sup>3</sup> | (Vacated) TWA: 5 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> |

| Component                                        | European Union | The United Kingdom                                                 | Germany                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------|----------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dibenzoyl peroxide                               |                | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr | TWA: 5 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>TWA: 4 mg/m <sup>3</sup> (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>TWA: 1 mg/m <sup>3</sup> (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>Höhepunkt: 8 mg/m <sup>3</sup><br>Höhepunkt: 4 mg/m <sup>3</sup> |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester |                | STEL: 15 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr |                                                                                                                                                                                                                                                                                                                             |

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

# SAFETY DATA SHEET

Perkadox CH-50

Revision Date 22-Mar-2025

|                                        |                                                                                                                                                                                                                               |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators                                                                                                               |
| <b>Recommended Filter type:</b>        | Particulates filter conforming to EN 143<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice                                                                                                                                                         |
| <b>Environmental exposure controls</b> | Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained                                                |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                                  |                          |                                          |
|--------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                                | White                    |                                          |
| <b>Physical State</b>                            | Solid                    |                                          |
| <b>Odor</b>                                      | Slight                   |                                          |
| <b>Odor Threshold</b>                            | No data available        |                                          |
| <b>pH</b>                                        | No information available |                                          |
| <b>Melting Point/Range</b>                       | No data available        |                                          |
| <b>Softening Point</b>                           | No data available        |                                          |
| <b>Boiling Point/Range</b>                       | No information available |                                          |
| <b>Flash Point</b>                               | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                          | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>                  | No information available |                                          |
| <b>Explosion Limits</b>                          | No data available        |                                          |
| <b>Vapor Pressure</b>                            | No data available        |                                          |
| <b>Vapor Density</b>                             | Not applicable           | Solid                                    |
| <b>Specific Gravity / Density</b>                | No data available        |                                          |
| <b>Bulk Density</b>                              | No data available        |                                          |
| <b>Water Solubility</b>                          | Insoluble                |                                          |
| <b>Solubility in other solvents</b>              | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b>   |                          |                                          |
| <b>Component</b>                                 | <b>log Pow</b>           |                                          |
| Dibenzoyl peroxide                               | 3.2                      |                                          |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 5.6                      |                                          |
| <b>Autoignition Temperature</b>                  | No data available        |                                          |
| <b>Decomposition Temperature</b>                 | No data available        |                                          |
| <b>Viscosity</b>                                 | Not applicable           | Solid                                    |
| <b>Explosive Properties</b>                      | No information available |                                          |
| <b>Oxidizing Properties</b>                      | No information available |                                          |
| <b>Molecular Formula</b>                         | C14 H10 O4               |                                          |
| <b>Molecular Weight</b>                          | 242.93                   |                                          |

# SAFETY DATA SHEET

Perkadox CH-50

Revision Date 22-Mar-2025

Self-Accelerating Decomposition Temperature (SADT) 55°C

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

Hazardous Polymerization No information available.  
Hazardous Reactions No information available.

### Conditions to Avoid

None known.

### Incompatible Materials

None known. Iron. Acids. Bases. Copper. Strong reducing agents.

### Hazardous Decomposition Products

benzoic acid. Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

#### (a) acute toxicity;

Oral Based on available data, the classification criteria are not met  
Dermal No data available  
Inhalation No data available

### Toxicology data for the components

| Component                                        | LD50 Oral                             | LD50 Dermal                           | LC50 Inhalation |
|--------------------------------------------------|---------------------------------------|---------------------------------------|-----------------|
| Dibenzoyl peroxide                               | LD50 = 7710 mg/kg ( Rat )             | -                                     | -               |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | LD50 > 2000 mg/kg ( Rat )<br>OECD 423 | LD50 > 2000 mg/kg ( Rat )<br>OECD 402 | -               |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 2

# SAFETY DATA SHEET

Perkadox CH-50

Revision Date 22-Mar-2025

**(d) respiratory or skin sensitization;**

**Respiratory**

No data available

**Skin**

Category 1

May cause sensitization by skin contact

**(e) germ cell mutagenicity;**

No data available

**(f) carcinogenicity;**

No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

Category 1B

**(h) STOT-single exposure;**

No data available

**(i) STOT-repeated exposure;**

No data available

**Target Organs**

No information available.

**(j) aspiration hazard;**

Not applicable

Solid

**Symptoms / effects, both acute and delayed**

Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

**Endocrine Disrupting Properties**

**Assess endocrine disrupting properties for human health**

Substance identified as having endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 Contains a substance on the National Authorities Endocrine Disruptor Lists

| Component                                                             | EU National Authorities Endocrine Disruptor Lists - Health |
|-----------------------------------------------------------------------|------------------------------------------------------------|
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester<br>84-61-7 ( 40-50 ) | List I List II                                             |

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects**

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                                        | Freshwater Fish                               | Water Flea                                  | Freshwater Algae                                              | Microtox                                        |
|--------------------------------------------------|-----------------------------------------------|---------------------------------------------|---------------------------------------------------------------|-------------------------------------------------|
| Dibenzoyl peroxide                               | LC50 = 0.06 mg/L 96h                          | EC50 = 0.11 mg/L 48h                        | EC50 = 0.06 mg/L 72h                                          |                                                 |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | LC50 > 2 mg/L Oryzias latipes, 96h (OECD 203) | EC50 > 2 mg/L Daphnia magna, 48h (OECD 202) | EC50 > 2 mg/L Psuedokirchneriella subcapitata, 72h (OECD 201) | NOEC > 100 mg/L activated sludge, 3h (OECD 209) |

**Persistence and degradability**

**Persistence**

Insoluble in water.

**Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

# SAFETY DATA SHEET

Perkadox CH-50

Revision Date 22-Mar-2025

**Bioaccumulative potential** May have some potential to bioaccumulate

| Component                                        | log Pow | Bioconcentration factor (BCF) |
|--------------------------------------------------|---------|-------------------------------|
| Dibenzoyl peroxide                               | 3.2     | No data available             |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 5.6     | No data available             |

**Mobility in soil** Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility.

## Endocrine Disruptor Information

| Component                                        | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances |
|--------------------------------------------------|------------------------------------------|--------------------------------------------------|
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | Group III Chemical                       |                                                  |

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

### Other Information

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Can be landfilled or incinerated, when in compliance with local regulations Do not let this chemical enter the environment Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN3106  
Hazard Class 5.2  
Proper Shipping Name ORGANIC PEROXIDE TYPE D, SOLID Dibenzoyl peroxide

### Road and Rail Transport

UN-No UN3106  
Hazard Class 5.2  
Proper Shipping Name ORGANIC PEROXIDE TYPE D, SOLID Dibenzoyl peroxide

### IATA

UN-No UN3106  
Hazard Class 5.2  
Proper Shipping Name ORGANIC PEROXIDE TYPE D, SOLID Dibenzoyl peroxide

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION



# SAFETY DATA SHEET

Perkadox CH-50

Revision Date 22-Mar-2025

## Safety, health and environmental regulations/legislation specific for the substance or mixture

### International Inventories

X = listed

| Component                                        | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Dibenzoyl peroxide                               | 202-327-6 | X    | X   | X     | X    | X    | X     | X    | KE-09889 |
| 1,2-Benzenedicarboxylic acid, dicyclohexyl ester | 201-545-9 | X    | X   | X     | X    | X    | X     | X    | KE-02215 |

### National Regulations

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

22-Mar-2025

Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

# SAFETY DATA SHEET

Perkadox CH-50

Revision Date 22-Mar-2025

---

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**